

# YUXUAN LI

**Email:** liyuxuan48@g.ucla.edu    **Website:** liyuxuan48.github.io    **GitHub:** github.com/liyuxuan48

## PROFILE

---

High-fidelity simulation has transformed thermal-fluid engineering, but its computational cost often limits its impact in practical design and optimization. I believe the next generation of simulation should be predictive, computationally efficient, and broadly accessible. My research seeks to bridge this gap through physics-based, reduced-order, and data-driven modeling for complex thermal-fluid systems, with applications in advanced thermal management and multiphase heat transfer.

## RESEARCH HIGHLIGHTS

---

- **Predictive Thermal-Fluid Modeling:** Developed experimentally validated, physics-based reduced-order models for oscillating heat pipe dynamics, thermal performance, and dryout.
- **Computational Methods for Heat Transfer:** Developed numerical modeling approaches for conjugate heat transfer, immersed boundary methods, and coupled fluid–solid thermal systems.
- **Industrial CFD and Multiphase Flow:** Built high-fidelity CFD models for cavitation, aeration, rotating machinery, and ship hydrodynamics, with emphasis on validation and engineering design workflows.
- **Reduced-Order Modeling of Complex Flows:** Developed Gaussian reduced-order models for late-time shock–bubble interaction and explored data-assimilation approaches for thermofluid parameter estimation.

## JOURNAL PUBLICATIONS

---

1. Yuxuan Li, Zachary Wong, Jeff D. Eldredge, Adrienne S. Lavine, Timothy S. Fisher, and Bruce L. Drolen. **“Effects of inclination angle on oscillating heat pipe thermal conductance and dryout.”** *Accepted for publication, ASME Journal of Heat and Mass Transfer*, 2026.
2. Yuxuan Li, Jeff D. Eldredge, Adrienne S. Lavine, Timothy S. Fisher, and Bruce L. Drolen. **“A Conjugate Heat Transfer Model of Oscillating Heat Pipe Dynamics, Performance, and Dryout.”** *International Journal of Heat and Mass Transfer* 227 (2024): 125530.
3. Li, Y., Z. Wang, B. Yu, B. Zhang, and H. Liu. **“Gaussian models for late-time evolution of two-dimensional shock–light cylindrical bubble interaction.”** *Shock Waves* 30, no. 2 (2020): 169–184.

## PROCEEDING PAPERS

---

1. Doruk Isik, Yuxuan Li, Azfar Ali, Todd Koehler, Chiranth Srinivasan. **“Numerical Approaches for Predicting Cavitation Inception in Non-Cavitating CFD Simulations.”** Proceedings of the ASME 2026 Fluids Engineering Division Summer Meeting, Bellevue, Washington (2026).
2. Yuxuan Li, Chengjie Wang. **“RANS-based simulation of KCS wave breaking characteristics.”** Wageningen 2025: A Workshop on CFD in Ship Hydrodynamics, Wageningen, Netherlands (2025).
3. Yuxuan Li, Jeff D. Eldredge, Adrienne S. Lavine, Timothy S. Fisher, and Bruce L. Drolen. **“A data assimilation model of oscillating heat pipe dynamics and performance.”** Joint 21st International Heat Pipe Conference and 15th International Heat Pipe Symposium, Melbourne, Australia (2023).

## PRESENTATIONS

---

1. Yuxuan Li, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **“Estimating thermofluid system parameters using a Markov chain Monte Carlo method, with an example of oscillating heat pipes.”** 76th Annual Meeting of the Division of Fluid Dynamics, Washington D.C., USA (2023).
2. Yuxuan Li, Zachary Wong, Jeff D. Eldredge, Timothy S. Fisher. **“Empirically Trained Models of Oscillating Heat Pipes for Improved Performance, Limits, and Control.”** 2023 Spacecraft Thermal Control Workshop, Torrance, USA (2023).
3. Li, Yuxuan, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **“Thermofluid modeling of an oscillating heat pipe.”** The 17th Southern California Flow Physics Symposium, San Diego, USA (2023).
4. Yuxuan Li, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **“Thermofluid modeling of an oscillating heat pipe.”** 75th Annual Meeting of the Division of Fluid Dynamics, Indianapolis, USA (2022).
5. Li, Yuxuan, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **“Thermofluid modeling of an oscillating heat pipe.”** The 16th Southern California Flow Physics Symposium, Los Angeles, USA (2022).

---

## APPOINTMENTS

### Simerics Inc., Novi, MI, USA

Project Engineer

2024–

- Developed high-fidelity multiphase CFD models using Simerics-MP+ for rotating machinery involving cavitation, aeration, and transient fluid dynamics.
- Performed transient simulations, validation against experiments, and parametric studies to improve predictive capability for industrial engineering applications.
- Collaborated with industrial partners to develop CFD workflows for engineering analysis, design optimization, and technical training.

---

## AWARDS

- UCLA Mechanical and Aerospace Engineering Department Fellowship 2019-2023
- SJTU Outstanding Graduate 2019
- UCLA-CSST Scholarship (~ 5000\$) 2018

---

## TEACHING EXPERIENCE

### University of California, Los Angeles

- Teaching Assistant for MAE150A *Intermediate Fluid Mechanics* (overall TA rating 7.85/9)

---

## PROFESSIONAL SERVICE

Reviewer

- ASME Journal of Heat and Mass Transfer
- International Journal of Thermal Sciences
- SAE WCX 2026

---

## EDUCATION

**University of California, Los Angeles**  
Ph.D. in Mechanical Engineering, Major in Fluid Mechanics

*2019–2024*  
GPA: 4.0/4.0

Thesis Advisors: Jeff D. Eldredge and Adrienne S. Lavine  
Research Mentors: Timothy S. Fisher and Bruce L. Drolen

**Shanghai Jiao Tong University**  
B.S. in Aeronautics and Astronautics

*2015–2019*  
GPA: 3.7/4.0